

News vendor Problem

The Fashion Store sells fashion items. The store has to order these items many months in advance of the fashion season in order to get a good price on the items. Each unit costs Fashion \$100. These units are sold to customers at a price of \$250 per unit. Items not sold during the season can be sold to the outlet store at \$80 per unit. If the store runs out of an item during the season it has to obtain the item from alternative sources and the cost including air freight to Fashion is \$190 per unit.

Fashion wants help in choosing the initial order quantity to maximize net contribution from running the store. What should we do?

86	94	90	86	82	84	91	76	85	83	92	82	89	88	79	83	83	85	89	90
73	84	86	90	90	92	83	91	85	85	82	81	81	76	81	81	78	85	84	82
88	86	85	88	86	89	87	84	83	79	90	87	83	87	82	81	85	84	87	89
82	80	92	85	88	85	83	87	84	84	86	80	87	80	89	79	83	80	86	85
81	93	91	89	80	86	87	86	88	84	81	84	84	82	77	93	94	97	87	75

Table 1: 100 demand data points for the Fashion Store.

Demand (D)	Frequency	P(Demand) $P(D)$	Cum. Prob. $P(\text{Demand} \leq D)$
73	1	0.01	0.01
75	1	0.01	0.02
76	2	0.02	0.04
77	1	0.01	0.05
78	1	0.01	0.06
79	3	0.03	0.09
80	5	0.05	0.14
81	7	0.07	0.21
82	7	0.07	0.28
83	8	0.08	0.36
84	10	0.10	0.46
85	10	0.10	0.56
86	9	0.09	0.65
87	8	0.08	0.73
88	5	0.05	0.78
89	6	0.06	0.84
90	5	0.05	0.89
91	3	0.03	0.92
92	3	0.03	0.95
93	2	0.02	0.97
94	2	0.02	0.99
97	1	0.01	1.00
	100	1.0	

Solution

Overage Cost (C_o) : $100 - 80 = 20$

Underage Cost (C_u): $(250-100)-(250-190) = 90$

$$P(D \leq Q^*) = \frac{C_u}{C_o + C_u} = \frac{90}{20+90} = 0.8182$$

Look into to frequency table, for service level of 0.8182. It shows Q^* is 89 units

Pat sells Sunday papers outside a retirement home on Sunday mornings. Pat collects \$1.75 per paper and pays \$.98 for a paper. The retirement home residents think fondly of Pat and aren't too upset when papers run out. On the other hand, any papers Pat is left with are thrown into the recycling bin. Pat is trying to figure out a systematic ordering policy. When Pat runs out of papers, the sale is lost.

Many Newsvendor problems are similar to this *lost sales* example, so we will define some notation that will help to generalize such examples.

r = revenue received for the sale of each newspaper = \$1.75

c = cost (wholesale) paid for each newspaper = \$.98

h = cost to hold any extra newspapers after the day is over = \$0

s = salvage value of a newspaper after the day is over or cost to dispose of a paper = \$0

(Note that s is positive if a salvage value and negative if a disposal cost.)

p = penalty for being unable to sell a customer a newspaper when one is desired = \$0:

includes loss-of-good will costs, backorder costs, emergency supply costs, etc.

- Overage Cost (C_o): $c + h - s$
- Underage Cost (C_u): $r - c + p$

$$P(D \leq Q^*) = \frac{C_u}{C_o + C_u} = \frac{(1.75 - 0.98 + 0)}{(0.98 + 0 - 0) + (1.75 - 0.98 + 0)} = 0.44$$

$$\text{Net Contribution } (Q) = \begin{cases} rD - cQ + s(Q - D) - h(Q - D); & \text{if } D \leq Q \\ (r - c)Q - p(D - Q); & \text{if } D \geq Q \end{cases}$$

- A newsstand owner is making a decision about how many copies of next day's newspaper should be ordered
- Each copy of newspaper costs \$0.5 and is sold for \$1
- The unsold newspapers at the end of the day will be returned to the publisher for a refund of \$0.25 per copy
- Assume that the number of copies sold during the day follows a uniform distribution $U(100, 200)$

What if the number of copies sold during the day follows a normal distribution $N(150, 20^2)$?

- Overage Cost (C_o): $c + h - s = 0.5 + 0 - 0.25 = 0.25$
- Underage Cost (C_u): $r - c + p:1 - 0.5 + 0 = 0.5$

$$P(D \leq Q^*) = \frac{C_u}{C_o + C_u} = \frac{0.5}{0.25 + 0.5} = 0.67 \text{ (app.)}$$

Case 1. Uniform Distribution

$$\frac{Q^* - 100}{200 - 100} = 0.67$$

$$Q^* = 167 \text{ (App.)}$$

Case 1. Normal Distribution

$$\frac{Q^* - 150}{20} = 0.43$$

$$Q^* = 159 \text{ (App.)}$$